



May 2024

Duration: 3 Hours

Max. Marks: 100

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

20 Marks

- 1-

11. The ratio of differential gain to the common mode gain for an Op-amp is called
 A) Slew Rate B) Input offset voltage
 C) CMRR D) PSRR
12. Which Flip-flop is available inside the IC 555?
 A) SR B) JK
 C) D D) T
13. In a feedback amplifier, if the feedback signal, V_f is out of phase with the input signal, V_{in} , then the feedback is called as
 A) positive feedback B) negative feedback
 C) open loop D) virtual ground
14. Which of the following feedback in a feedback circuit, leads to oscillations?
 A) positive feedback B) negative feedback
 C) Neutral feedback D) both positive and negative feedback
15. In a RC-phase shift oscillator, each RC section is designed for a phase shift of _____ degree.
 A) 30 B) 60
 C) 90 D) 180
16. Which of the following oscillator is used for oscillations at high frequency range from 100 KHz & above to GHz?
 A) RC phase (lead) shift B) RC phase (lag) shift
 C) Colpitts D) Moderate frequency
17. Which of the following type of medium is cheaper and more immune to interference and noise?
 A) Copper cable B) Silver cable
 C) Co-axial cable D) Optical fibres
18. _____ is basically an unwanted signal that disturbs the communication system.
 A) Voltage signal B) Current signal
 C) Noise signal D) Spike signal
19. Microprocessors/controllers based on the _____ architecture will have separate data bus and instruction bus.
 A) Von-Neumann B) Canadian
 C) Oxford D) Harvard
20. Which of the following is a form of transducer device (mechanical or electrical) which converts signals to corresponding physical action?
 A) Sensor B) Motor
 C) Actuator D) Amplifier

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Compare various pn-junction diode equivalent circuits with VI-characteristics.	6	L*2	1	1
	b) Derive expressions for the following in the case of a half-wave rectifier. i) Ripple factor ii) Rectification efficiency.	5	L2	1	1
	c) A battery source of 5V is connected in series with a resistor of 100 Ω and a Silicon pn-junction diode. Write the circuit. Determine the diode forward current.	5	L3	1	1
2.	a) With a neat circuit diagram and waveform, explain the operation of a bridge rectifier.	6	L2	1	1
	b) How does the capacitor filter connected across the load resistor in a full wave rectifier reduces the AC component at the output terminals? Explain with the circuit and waveforms.	5	L2	1	1
	c) A 24V, 600mW Zener diode is used for providing a 24V stabilized DC supply to a variable load from a 32V supply. Calculate, (i) Value of series resistor required (ii) Zener current when the load resistance is 1200 Ω .	5	L3	1	1

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| 3. a) | Write the circuit of a transistor in Common Emitter configuration and mention the need for base and collector resistors. Explain the input and output characteristics. | 6 | L2 | 2 | 1 |
| b) | A transistor has a collector current of 3mA, emitter current of 3.03mA. Find the current gain in CE mode (β) for the transistor. Assuming no change in the base current, find the new collector current when the transistor is replaced with a new one with β of 70. | 5 | L3 | 2 | 1 |
| c) | Explain the drain and transfer characteristics of JFET. Write the equation that relates the drain current and gate-source voltage. | 5 | L2 | 2 | 1 |

Unit – II

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| 4. a) | With a neat circuit diagram, explain the application of Operational amplifier (Op-amp) as an Integrator. Derive the expression for output voltage. | 6 | L2 | 3 | 1 |
| b) | Write the characteristics of an ideal Op-amp. | 5 | L1 | 3 | 1 |
| c) | Design an inverting adder circuit using Op-amp to obtain the output voltage given by, $V_o = 2[0.1V_1 + 0.5V_2 + 2V_3]$. Here, V_1 , V_2 and V_3 are the input voltages. Assume a feedback resistor of 10K Ω . Write the final circuit. | 5 | L3 | 3 | 1 |
| 5. a) | Explain the application of IC 555 Timer as an astable multivibrator. Write the expression for frequency of oscillation. | 6 | L2 | 3 | 1 |
| b) | Explain the operation of non-inverting comparator with positive voltage. Draw the necessary diagram and waveforms. | 5 | L2 | 3 | 1 |
| c) | For an IC 555 based timer circuit as an astable multivibrator, the duty cycle = 75%, frequency of operation = 1KHz, $R_2 = 3.6K\Omega$, $C = 0.1\mu F$, then determine R_1 . | 5 | L3 | 3 | 1 |
| 6. a) | With a block diagram, state explain Barkhausen's criteria for sustained oscillations. Derive the condition used. | 6 | L2 | 4 | 1 |
| b) | With a neat circuit diagram and relevant equations, explain the principle of operation of Hartley Oscillator. | 5 | L2 | 4 | 1 |
| c) | In a Colpitts oscillator, $C_1 = 100pF$, $C_2 = 260pF$. Find the value of the inductor L, if the frequency of oscillation is 40KHz. | 5 | L3 | 4 | 1 |

Unit – III

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| 7. a) | With a block diagram, explain a communication system. | 6 | L2 | 5 | 1 |
| b) | Explain the capacity of a cell with the necessary equations | 5 | L2 | 5 | 1 |
| c) | What are the effects of interference in a cellular system? Discuss the type interferences in a cellular region. | 5 | L2 | 5 | 1 |
| 8. a) | With a block diagram, explain various elements of an embedded system. | 6 | L2 | 5 | 1 |
| b) | Distinguish between Harvard and Von-Neumann Architecture of embedded system. | 5 | L2 | 5 | 1 |
| c) | Discuss an optocoupler with a diagram. Explain how it can be interfaced with the microcontroller. | 5 | L2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
